

INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH03

Unit 3: Inorganic 2 and Physical 2

Mark scheme

January 2025

Version: 1.0 Final



2 5 1 X C H 0 3 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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A-level Chemistry

Mark Scheme Instructions for Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the examiner make his or her judgement and help to delineate what is acceptable or not worthy of credit or, in discursive answers, to give an overview of the area in which a mark or marks may be awarded.

The extra information in the 'Comments' column is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

You should mark according to the contents of the mark scheme. If you are in any doubt about applying the mark scheme to a particular response, consult your Team Leader.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which might confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

The use of M1, M2, M3 etc in the right-hand column refers to the marking points in the order in which they appear in the mark scheme. So, M1 refers to the first marking point, M2 the second marking point etc.

2. Emboldening

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **OR**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general 'List' principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of error / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (often prefaced by 'Ignore' in the mark scheme) are not penalised.

For example, in a question requiring 2 answers for 2 marks:

Correct answers	Incorrect answers (ie incorrect rather than neutral)	Mark (2)	Comment
1	0	1	
1	1	1	They have not exceeded the maximum number of responses so there is no penalty.
1	2	0	They have exceeded the maximum number of responses so the extra incorrect response cancels the correct one.
2	0	2	
2	1	1	
2	2	0	
3	0	2	The maximum mark is 2
3	1	1	The incorrect response cancels out one of the two correct responses that gained credit.
3	2	0	Two incorrect responses cancel out the two marks gained.
3	3	0	

3.2 Marking procedure for calculations

Full marks should be awarded for a correct numerical answer, without any working shown, unless the question states 'Show your working' or 'justify your answer'. In this case, the mark scheme will clearly indicate what is required to gain full credit.

If an answer to a calculation is incorrect and working is shown, process mark(s) can usually be gained by correct substitution / working and this is shown in the 'Comments' column or by each stage of a longer calculation.

3.3 Errors carried forward, consequential marking and arithmetic errors

Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ECF or consequential in the marking scheme.

An arithmetic error should be penalised for one mark only unless otherwise amplified in the marking scheme. Arithmetic errors may arise from a slip in a calculation or from an incorrect transfer of a numerical value from data given in a question.

3.4 Equations

In questions requiring students to write equations, state symbols are generally ignored unless otherwise stated in the 'Comments' column.

Examiners should also credit correct equations using multiples and fractions unless otherwise stated in the 'Comments' column.

3.5 Oxidation states

In general, the sign for an oxidation state will be assumed to be positive unless specifically shown to be negative.

3.6 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.7 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term or if the question requires correct IUPAC nomenclature.

3.8 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.9 Ignore / Insufficient / Do not allow

Ignore or insufficient is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

Do **not** allow means that this is a wrong answer which, even if the correct answer is given, will still mean that the mark is not awarded.

3.10 Marking crossed out work

Crossed out work that **has not been** replaced should be marked as if it were not crossed out, if possible. Where crossed out work **has been** replaced, the replacement work and not the crossed out work should be marked.

3.11 Reagents

The command word “Identify”, allows the student to choose to use **either** the name or the formula of a reagent in their answer. In some circumstances, the list principle may apply when both the name and the formula are used. Specific details will be given in mark schemes.

The guiding principle is that a reagent is a chemical which can be taken out of a bottle or container. Failure to identify complete reagents **will be penalised**, but follow-on marks (eg for a subsequent equation or observation) can be scored from an incorrect attempt (possibly an incomplete reagent) at the correct reagent. Specific details will be given in mark schemes.

For example, **no credit** would be given for:

- the cyanide ion or CN^- when the reagent should be potassium cyanide or KCN
- the hydroxide ion or OH^- when the reagent should be sodium hydroxide or NaOH
- the $\text{Ag}(\text{NH}_3)_2^+$ ion when the reagent should be Tollens' reagent (or ammoniacal silver nitrate). In this example, no credit is given for the ion, but credit could be given for a correct observation following on from the use of the ion. Specific details will be given in mark schemes.

In the event that a student provides, for example, **both** KCN and cyanide ion, it would be usual to ignore the reference to the cyanide ion (because this is not contradictory) and credit the KCN. Specific details will be given in mark schemes.

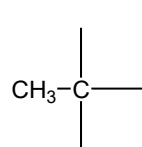
3.12 Organic structures

Where students are asked to draw organic structures, unless a specific type is required in the question and stated in the mark scheme, these may be given as displayed, structural or skeletal formulas or a combination of all three as long as the result is unambiguous.

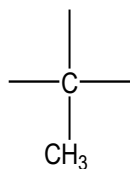
In general

- Displayed formulae must show all of the bonds and all of the atoms in the molecule, but need not show correct bond angles.
- Skeletal formulae must show carbon atoms by an angle or suitable intersection in the skeleton chain. Functional groups must be shown and it is essential that all atoms other than C atoms are shown in these (except H atoms in the functional groups of aldehydes, secondary amines and N-substituted amides which do not need to be shown).
- Structures must not be ambiguous, eg 1-bromopropane should be shown as $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and not as the molecular formula $\text{C}_3\text{H}_7\text{Br}$ which could also represent the isomeric 2-bromopropane.
- Bonds should be drawn correctly between the relevant atoms. This principle applies in all cases where the attached functional group contains a carbon atom, eg nitrile, carboxylic acid, aldehyde and acid chloride. The carbon-carbon bond should be clearly shown. Wrongly bonded atoms will be penalised on every occasion. (see the examples below)
- The same principle should also be applied to the structure of alcohols. For example, if students show the alcohol functional group as $\text{C} - \text{HO}$, they should be penalised on every occasion.
- Latitude should be given to the representation of $\text{C} - \text{C}$ bonds in alkyl groups, given that CH_3- is considered to be interchangeable with $\text{H}_3\text{C}-$ even though the latter would be preferred.
- Similar latitude should be given to the representation of amines where NH_2- C will be allowed, although $\text{H}_2\text{N}-$ C would be preferred.
- Poor presentation of vertical $\text{C} - \text{CH}_3$ bonds or vertical $\text{C} - \text{NH}_2$ bonds should not be penalised. For other functional groups, such as $-\text{OH}$ and $-\text{CN}$, the limit of tolerance is the half-way position between the vertical bond and the relevant atoms in the attached group.

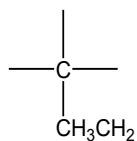
By way of illustration, the following would apply.



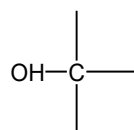
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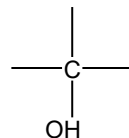
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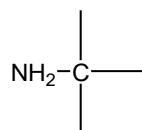
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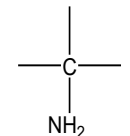
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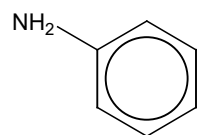
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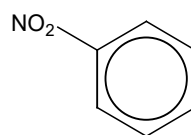
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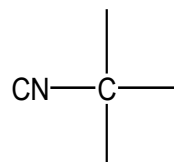
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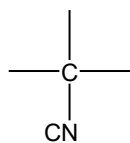
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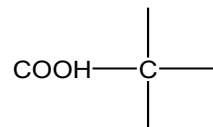
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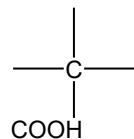
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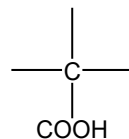
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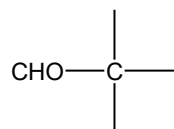
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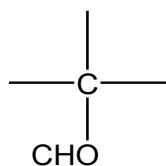
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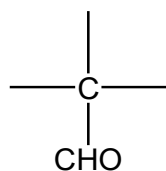
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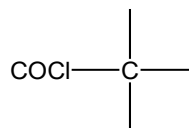
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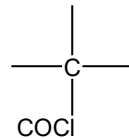
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- Representation of CH_2 by C-H_2 will be penalised.
- Some examples are given here of **structures** for specific compounds that should **not** gain credit (but, exceptions may be made in the context of balancing equations).

CH_3COH for ethanal

$\text{CH}_3\text{CH}_2\text{HO}$ for ethanol

OHCH_2CH_3 for ethanol

$\text{C}_2\text{H}_6\text{O}$ for ethanol

CH_2CH_2 for ethene

$\text{CH}_2.\text{CH}_2$ for ethene

$\text{CH}_2:\text{CH}_2$ for ethene

- Each of the following **should gain credit** as alternatives to correct representations of the structures.

$\text{CH}_2 = \text{CH}_2$ for ethene, $\text{H}_2\text{C}=\text{CH}_2$

$\text{CH}_3\text{CHOHCH}_3$ for propan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$

- In most cases, the use of “sticks” to represent C-H bonds in a structure should **not** be penalised. The exceptions to this when “sticks” will be penalised include:
 - structures in mechanisms where the C-H bond is essential (eg elimination reactions in halogenoalkanes and alcohols)
 - when a displayed formula is required
 - when a skeletal structure is required or has been drawn by the candidate.

3.13 Organic names

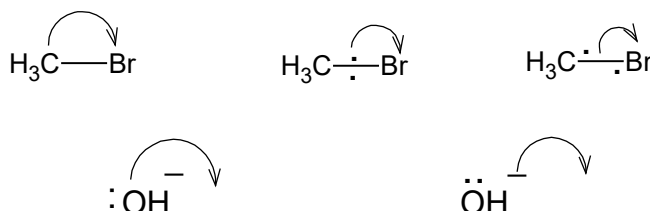
As a general principle, non-IUPAC names or incorrect spelling or incomplete names should **not** gain credit. Some illustrations are given here.

but-2-ol	should be butan-2-ol
2-hydroxybutane	should be butan-2-ol
butane-2-ol	should be butan-2-ol
2-butanol	should be butan-2-ol
ethan-1,2-diol	should be ethane-1,2-diol
2-methpropan-2-ol	should be 2-methylpropan-2-ol
2-methylbutan-3-ol	should be 3-methylbutan-2-ol
3-methylpentan	should be 3-methylpentane
3-mythylpentane	should be 3-methylpentane
3-methypentane	should be 3-methylpentane
propanitrile	should be propanenitrile
aminethane	should be ethylamine (although aminoethane can gain credit)
2-methyl-3-bromobutane	should be 2-bromo-3-methylbutane
3-bromo-2-methylbutane	should be 2-bromo-3-methylbutane
3-methyl-2-bromobutane	should be 2-bromo-3-methylbutane
2-methylbut-3-ene	should be 3-methylbut-1-ene
difluorodichloromethane	should be dichlorodifluoromethane

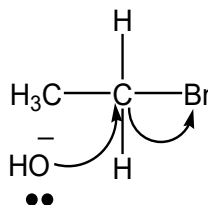
3.14 Organic reaction mechanisms

Curly arrows should originate either from a lone pair of electrons or from a bond.

The following representations should not gain credit **and will be penalised each time** within a clip.



For example, the following would score zero marks



When the curly arrow is showing the formation of a bond to an atom, the arrow can go directly to the relevant atom, alongside the relevant atom or **more than half-way** towards the relevant atom.

In free-radical substitution:

- the absence of a radical dot should be penalised **once only** within a clip
- the use of half-headed arrows is not required, but the use of double-headed arrows or the incorrect use of half-headed arrows in free-radical mechanisms should be penalised **once only** within a clip.

The correct use of skeletal formulae in mechanisms is acceptable, but where a C–H bond breaks both the bond and the H must be drawn to gain credit.

Question	Marking guidance	Mark	Comments
01.1	X: $\text{Ca}^{2+}(\text{g})$ Y: e^{-}	1	state symbol essential both needed for mark

Question	Marking guidance	Mark	Comments
01.2	A: (first) electron affinity of oxygen	1	
	B: <u>lattice dissociation</u> enthalpy of calcium oxide	1	

Question	Marking guidance	Mark	Comments
01.3	$\Delta_f H^\ominus(\text{CaO}) + 3513 = 193 + 248 + 590 + 1150 - 142 + 844$	1	+630 scores 1 mark
	$\Delta_f H^\ominus(\text{CaO}) = -630 \text{ (kJ mol}^{-1}\text{)}$	1	

Total		5	
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Question	Marking guidance	Mark	Comments
02.1	$q = mc\Delta T = 25.0 \times 4.18 \times 14.2$	1	
	$q = 1483.9 \text{ (J)}$	1	
	$n(\text{MgCl}_2) = \frac{1.57}{95.3} = 0.0165 \text{ (mol)}$		
	$\Delta H = \frac{-q}{n} = \frac{-1.484}{0.0165}$	1	$\text{M3} = \frac{-\text{M1} \times 10^{-3}}{\text{M2}}$ $\Delta H = -89.9 \text{ (kJ mol}^{-1}\text{)}$ M3 answer must be negative

Question	Marking guidance		Comments
02.2	magnesium hydroxide is insoluble (in cold water) or magnesium hydroxide has very low solubility (in cold water)	1	allow the magnesium hydroxide would not dissolve in 25.0 cm ³ of water allow magnesium hydroxide is sparingly soluble

Question	Marking guidance	Mark	Comments
02.3	$\Delta_{\text{sol}}H (\text{MgCl}_2) = 2493 - 1920 - 2(364)$	1	
	$\Delta_{\text{sol}}H (\text{MgCl}_2) = -155 \text{ (kJ mol}^{-1}\text{)}$	1	

Total		6	
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Question	Marking guidance	Mark	Comments
03.1	the molecules are stationary or there is no disorder	1	

Question	Marking guidance	Mark	Comments
03.2	data read from the graph at 373K (maximum entropy value – minimum entropy value) $132 - 24 = (+) 108 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}$	1	

Question	Marking guidance	Mark	Comments
03.3	$\Delta S = \frac{\Delta H}{T}$ or $\Delta H = \Delta S \times T$ $\Delta H = \frac{108}{1000} \times 373$ $\Delta H = 40.3 \text{ (kJ mol}^{-1}\text{)}$	1	
		1	$\Delta H = \frac{\text{ans to Qu 03.2}}{1000} \times 373$ using alternative answer $\Delta H = \frac{65.0}{1000} \times 373$ $\Delta H = 24.2 \text{ (kJ mol}^{-1}\text{)}$

Question	Marking guidance	Mark	Comments
03.4	change in disorder is larger at boiling (than melting)	1	

Total		5	
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Question	Marking guidance	Mark	Comments
04.1	$n(\text{Na}_2\text{C}_2\text{O}_4) = \frac{0.910}{134} = 6.79 \times 10^{-3} \text{ (mol)}$	1	
	$[\text{Na}_2\text{C}_2\text{O}_4] = \frac{6.79 \times 10^{-3} \times 1000}{150} = 0.0453 \text{ (mol dm}^{-3}\text{)}$	1	$M2 = M1 \times \frac{1000}{150}$
	$n(\text{Na}_2\text{C}_2\text{O}_4) = \frac{0.0453 \times 25.0}{1000} = 1.13 \times 10^{-3} \text{ (mol)}$	1	$M3 = M2 \times 25 \times 10^{-3}$ or $M3 = M1 \times \frac{25}{150}$
	$n(\text{MnO}_4^-) = 1.13 \times 10^{-3} \times \frac{2}{5} = 4.53 \times 10^{-4} \text{ (mol)}$	1	$M4 = M3 \times \frac{2}{5}$
	$[\text{MnO}_4^-] = \frac{4.53 \times 10^{-4} \times 1000}{22.35} = 0.0203 \text{ (mol dm}^{-3}\text{)}$	1	$M5 = M4 \times \frac{1000}{22.35}$

Question	Marking guidance	Mark	Comments
04.2	colourless to (pale) pink	1	do not accept colourless to purple

Question	Marking guidance	Mark	Comments
04.3	MnO_4^- and $\text{C}_2\text{O}_4^{2-}$ ions repel or both ions are negative so they repel	1	allow high activation energy

Question	Marking guidance	Mark	Comments
04.4	$4 \text{Mn}^{2+} + \text{MnO}_4^- + 8 \text{H}^+ \rightarrow 5 \text{Mn}^{3+} + 4 \text{H}_2\text{O}$	1	allow equations in either order
	$2 \text{Mn}^{3+} + \text{C}_2\text{O}_4^{2-} \rightarrow 2 \text{CO}_2 + 2 \text{Mn}^{2+}$	1	allow multiples ignore state symbols
Total		9	

Question	Marking guidance	Mark	Comments
05.1	$n(\text{H}^+) = \frac{25 \times 0.150}{1000} = 3.75 \times 10^{-3} \text{ (mol)}$	1	
	$[\text{H}^+] = \frac{1000 \times 3.75 \times 10^{-3}}{40.0} = 0.09375 \text{ (mol dm}^{-3}\text{)}$	1	M2 = M1 $\times$ 25
	$\text{pH} = -\log[\text{H}^+] = 1.03$	1	M3 = $-\log(\text{M2})$ if there has been an attempt to calculate a concentration of H^+ alternative method: M1: dilution ratio = $\frac{25.0}{40.0} = 0.625$ M2 = alternative M1 $\times$ 0.150 = 0.09375 (mol dm ⁻³) M3 = $-\log(\text{M2})$ if there has been an attempt to calculate a concentration of H^+

Question	Marking guidance	Mark	Comments
05.2	$[\text{OH}^-] = 0.650 \times 2 = 1.30 \text{ (mol dm}^{-3}\text{)}$ $[\text{H}^+] = \frac{1.00 \times 10^{-14}}{1.30} = 7.69 \times 10^{-15} \text{ (mol dm}^{-3}\text{)}$ $\text{pH} = -\log [\text{H}^+] = 14.11$	1	
		1	$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{\text{M1}}$
		1	M3 = $-\log(\text{M2})$ if there has been an attempt to calculate a concentration of H^+ using K_w $\text{pH} = 13.81$ scores 2 alternative M2: $\text{pOH} = -\log[\text{OH}^-] = -0.11(39)$ M3: $\text{pH} = 14 - \text{pOH} = 14 - (-0.11) = 14.11$

Question	Marking guidance	Mark	Comments
05.3	remains constant	1	

Question	Marking guidance	Mark	Comments
05.4	$[H^+] = 10^{-pH} = 10^{-4.50} = 3.16 \times 10^{-5} \text{ (mol dm}^{-3}\text{)}$	1	
	$[A^-] = K_a \frac{[HA]}{[H^+]} = 1.76 \times 10^{-5} \times \frac{0.15}{3.16 \times 10^{-5}} = 0.0835 \text{ (mol dm}^{-3}\text{)}$	1	M2 = $K_a \times \frac{0.15}{M1}$
	$n(\text{ethanoate}) = 0.0835 \times \frac{25.0}{1000} = 2.09 \times 10^{-3} \text{ (mol)}$	1	M3 = $M2 \times 25.0 \times 10^{-3}$
	$\text{mass} = 2.09 \times 10^{-3} \times 82.0 = 0.171 \text{ (g)}$	1	M4 = $M3 \times 82.0$ Alternative method using Henderson-Hasselbalch equation: M1: $pK_a = -\log K_a = 4.75$ $pH = pK_a + \log \frac{[A^-]}{[HA]}$ M2: $[A^-] = [HA] \times 10^{(pH-pK_a)} = 0.0835 \text{ (mol dm}^{-3}\text{)}$
Total		11	

Question	Marking guidance	Mark	Comments
06.1	name of A: Salt bridge	1	
	purpose: completes the circuit	1	allow it allows ions to move or to maintain electrical neutrality

Question	Marking guidance	Mark	Comments
06.2	platinum (black)	1	

Question	Marking guidance			Mark	Comments
06.3					ignore 298K
		Identity	Condition		
	C	HCl	1.00 mol dm ⁻³	1	allow 0.50 mol dm ⁻³ H ₂ SO ₄ or
	D	H ₂	100 kPa	1	1.00 mol dm ⁻³ HNO ₃ not 1 atm

Question	Marking guidance	Mark	Comments
06.4	use $2.00 \text{ mol dm}^{-3} \text{ K}_2\text{Cr}_2\text{O}_7$ (and $1.00 \text{ mol dm}^{-3} \text{ Cr}_2(\text{SO}_4)_3$)	1	Mark independently
	so $[\text{Cr}_2\text{O}_7^{2-}]$ is 1.00 mol dm^{-3} (and $[\text{Cr}^{3+}]$ is 1.00 mol dm^{-3})	1	

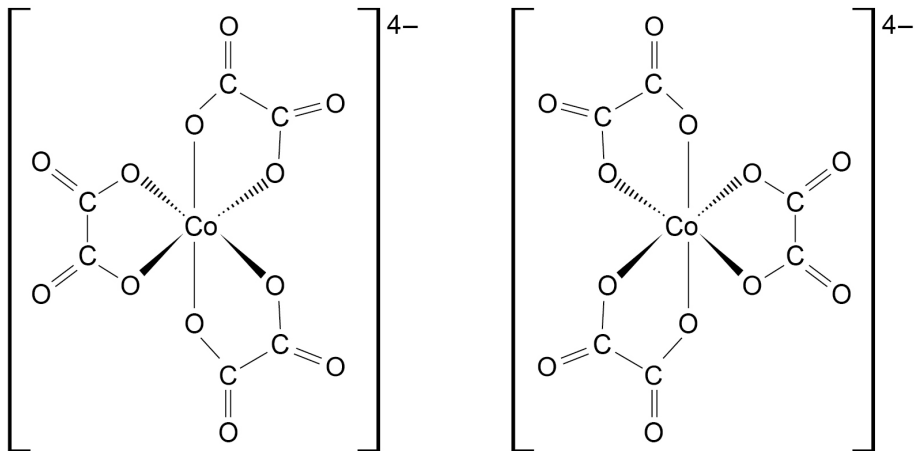
Question	Marking guidance	Mark	Comments
06.5	$\text{Pt} \mid \text{H}_2 \mid \text{H}^+ \parallel$	1	LHS
	$\text{Cr}_2\text{O}_7^{2-}, (\text{H}^+), \text{Cr}^{3+} \mid \text{Pt}$	1	RHS allow 1 mark for correct cell reversed allow 1 mark for correct cell with Pt electrodes missing.

Question	Marking guidance	Mark	Comments
06.6	$\text{Cr}_2\text{O}_7^{2-} + 8\text{H}^+ + 3\text{H}_2 \rightarrow 7\text{H}_2\text{O} + 2\text{Cr}^{3+}$	1	allow multiples ignore state symbols

Question	Marking guidance	Mark	Comments
06.7	(+)1.36 V	1	

Total		11	
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Question	Marking guidance	Mark	Comments
07.1	(a bidentate ligand) contains <u>two atoms</u> which each donate a lone pair of electrons (to a central metal ion) or (a bidentate ligand) contains <u>two atoms</u> which each form a co-ordinate bond (to a central metal ion)	1	

Question	Marking guidance	Mark	Comments
07.2		3	M1 for correct displayed formula of one complex with three ethanedioate ligands. M2 for a correct mirror image of the complex with three ethanedioate ligands M3 charges on each cobalt complex.

Question	Marking guidance	Mark	Comments
07.3	$[\text{Co}(\text{H}_2\text{O})_6]^{2+} + 3 \text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2 \rightarrow$ $[\text{Co}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+} + 6 \text{H}_2\text{O}$	1	
	the reaction goes from 4 moles on the LHS to 7 moles on the RHS / there is an increase in the number of particles.	1	
	(so) disorder / entropy increases / ΔS is positive	1	
	the same number of bonds are broken and formed and Co–O and Co–N have similar bond energies	1	
	(therefore) the enthalpy change is (approximately) zero / very small	1	
	so the free-energy change is less than or equal to zero / $\Delta G \leq 0$	1	
Total		10	

Question	Marking guidance	Mark	Comments
08.1	$\text{P}_4(\text{s}) + 5 \text{O}_2(\text{g}) \rightarrow \text{P}_4\text{O}_{10}(\text{s})$	1	allow equations using P rather than P_4 do not allow P_2O_5
	$\text{P}_4(\text{s}) + 10 \text{Cl}_2(\text{g}) \rightarrow 4 \text{PCl}_5(\text{s})$	1	must have state symbols allow 1 mark if both equations are correct but state symbols are missing / incorrect
	(bright)white flame or white solid / ash	1	do not allow smoke / precipitate

Question	Marking guidance	Mark	Comments
08.2	$\text{Al}_2\text{O}_3 + 6 \text{HCl} + 9 \text{H}_2\text{O} \rightarrow 2 [\text{Al}(\text{H}_2\text{O})_6]^{3+} + 6 \text{Cl}^-$ or $\text{Al}_2\text{O}_3 + 6 \text{H}^+ + 9 \text{H}_2\text{O} \rightarrow 2 [\text{Al}(\text{H}_2\text{O})_6]^{3+}$	1	allow $\text{Al}_2\text{O}_3 + 6 \text{HCl} \rightarrow 2 \text{AlCl}_3 + 3 \text{H}_2\text{O}$ $\text{Al}_2\text{O}_3 + 6 \text{H}^+ \rightarrow 2 \text{Al}^{3+} + 3 \text{H}_2\text{O}$

Question	Marking guidance	Mark	Comments
08.3	no visible change/reaction or white solid remains	1	

Question	Marking guidance	Mark	Comments
08.4	$3\text{Ba}(\text{OH})_2 + 2\text{H}_3\text{PO}_4 \rightarrow \text{Ba}_3(\text{PO}_4)_2 + 6\text{H}_2\text{O}$	1	allow $3\text{Ba}^{2+} + 2\text{PO}_4^{3-} \rightarrow \text{Ba}_3(\text{PO}_4)_2$
	white precipitate	1	

Total		7	
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Question	Marking guidance	Mark	Comments
09.1	Cu ²⁺ absorbs some visible / red wavelengths of light when d-orbital electrons are excited	1	do not allow emitted
	(so the solution) transmits / reflects blue wavelengths of light	1	

Question	Marking guidance	Mark	Comments
09.2	$\lambda = \frac{hc}{E}$	1	
	$\lambda = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{3.79 \times 10^{-19}} (= 5.25 \times 10^{-7} \text{ m})$ 525 nm	1	

Question	Marking guidance	Mark	Comments
09.3	Observations (blue solution to) blue precipitate (which reacts to give a) dark blue solution	1	
		1	
	Equations $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 2\text{NH}_3 \rightarrow \text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2 + 2\text{NH}_4^+$	1	
	$\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 2\text{OH}^- + 2\text{H}_2\text{O}$	1	allow $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 4\text{H}_2\text{O}$
Total		8	

Question	Marking guidance	Mark	Comments
10.1	M1 expression $K_a = \frac{[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}][\text{H}^+]}{[\text{Fe}(\text{H}_2\text{O})_6]^{3+}}$	1	do not insist on square brackets as part of the complex, but must have square brackets for the expression.
	M2 K_a from $\text{p}K_a$ $K_a = 10^{-2.22} (= 6.0256 \times 10^{-3} \text{ mol dm}^{-3})$	1	
	M3 amount of Fe^{3+} in moles $n(\text{Fe}^{3+}) = \frac{1.35}{162.3} = 8.32 \times 10^{-3} \text{ (mol)}$	1	
	M4 concentration of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} = \frac{8.32 \times 10^{-3} \times 1000}{750} = 0.0111 \text{ (mol dm}^{-3}\text{)}$	1	$\text{M4} = \frac{\text{M3} \times 1000}{750}$
	M5 calculation of $[\text{H}^+]$ $([\text{H}^+] = \sqrt{K_a[\text{Fe}(\text{H}_2\text{O})_6]^{3+}})$ $[\text{H}^+] = \sqrt{6.0256 \times 10^{-3} \times 0.0111} = 8.17 \times 10^{-3} \text{ (mol dm}^{-3}\text{)}$	1	$\text{M5} = \sqrt{\text{M2} \times \text{M4}}$
	M6 calculation of pH $\text{pH} = -\log [\text{H}^+]$ $\text{pH} = 2.09$	1	$\text{M6} = -\log \text{M5}$

Question	Marking guidance	Mark	Comments
10.2	Fe^{2+} has a lower charge (density) than Fe^{3+}	1	Al^{3+} has a higher charge density (than Fe^{3+})
	Al^{3+} has a (much) smaller (ionic) radius (than Fe^{3+} but the same charge)	1	
Total		8	